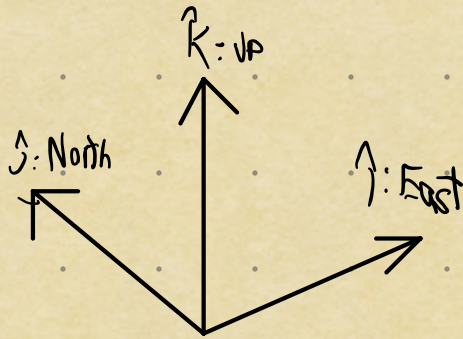


Navigation frame:

Fixed frame relative to ground; for our purpose it is inertial. Origin at launchpad

Axes:

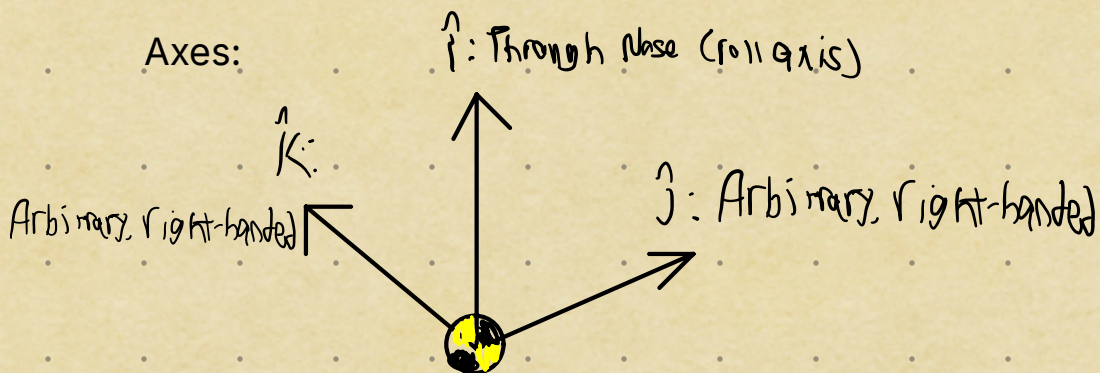


Notate Vectors as v_N

Body Frame:

Frame attached to airframe, origin at center of mass at launch

Axes:



Notate vectors as v_B

Notate matrices as R_{XY} , Y is starting frame and X is ending frame

Directional Cosine Matrix (DCM)

$R[i][j]$ is the cosine of angle between axis i in one frame and j in another

Because both coordinate systems are orthonormal, some properties:

$$R * \text{transpose}(R) = I$$

$$\det(R) = 1$$

$$\text{Inverse} = \text{transpose}$$

3 elementary rotations in body frame:

About x (ϕ , roll)

About y (θ , pitch)

About z (ψ , yaw)

Each rotation can be represent with a matrix; product of all 3 gives DCM matrix

ORDER MATTERS; for body to nav mapping, **ALWAYS DO IN THIS ORDER:**

$$\mathbf{DCM} = \mathbf{R}_x * \mathbf{R}_y * \mathbf{R}_z$$

Gimbal Lock:

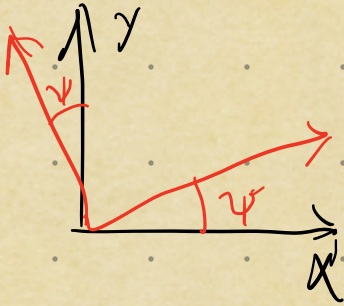
When 2 axes become aligned (usually when one angle is 90 deg), you use the ability to freely rotate one of them

Euler to DCM

Yaw: Abt. z



$$\hat{i} \rightarrow \begin{pmatrix} \cos \psi \\ \sin \psi \\ 0 \end{pmatrix}$$

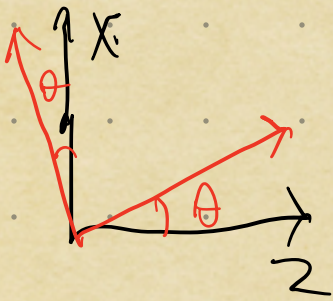


$$\hat{j} \rightarrow \begin{pmatrix} -\sin \psi \\ \cos \psi \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\hat{k} \rightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Pitch: Abt. y



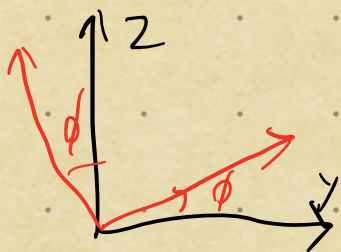
$$\hat{i} \rightarrow \begin{pmatrix} \cos \theta \\ 0 \\ -\sin \theta \end{pmatrix}$$

$$\hat{j} \rightarrow \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\hat{k} \rightarrow \begin{pmatrix} \sin \theta \\ 0 \\ \cos \theta \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$$

Roll: Abt. x



$$\hat{i} \rightarrow \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\hat{j} \rightarrow \begin{pmatrix} 0 \\ \cos \phi \\ \sin \phi \end{pmatrix}$$

$$\hat{k} \rightarrow \begin{pmatrix} 0 \\ -\sin \phi \\ \cos \phi \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{pmatrix}$$

↑
Passive
rots,

↑
Active rots

We want passive rots, so transverse all margins
manipulate